Polish-Japanese Academy of Information Technology

Bachelor's engineering thesis

Short-Term Forecasting of Martian Surface Weather Using Machine Learning

PRO3, Task 04: Chapter 4, Data exploration, standardization and normalization

Uladzislau Staliarou (s29905)

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4 Data exploration, standardization and normalization

The station records described in Chapter 3 cannot be passed to a forecasting model as they are. They come from three instruments with different sampling schemes, clocks and missing-value codes. They contain gaps of every length, from seconds to months, and a few variables carry values that no sensor reading can justify. Their largest variations, the diurnal and seasonal cycles and the slow drift of a moving rover, are also much larger than the short-term changes that the models are meant to forecast. This chapter describes how the data were explored, standardized and normalized before modeling.

The three terms are used here in the following sense. Exploration, or exploratory data analysis, is the systematic inspection of the structure, distributions, relations and anomalies of a data set, mainly through summary statistics and graphics, before any model is fitted[1]. Standardization brings the records into one consistent form: common variable names and units, one representation of missing values, one time base and common time grids. Normalization transforms the values themselves to a common scale or reference, for example by rescaling them, by removing a seasonal baseline or by expressing them relative to their recent past. Rescaling to zero mean and unit variance is itself often called standardization; in this chapter it is treated as a normalization technique. All data of the project are tabular time series (Chapter 3), so the methods specific to text, images, video and audio do not apply.

Section 4.1 presents the exploration, Section 4.2 the standardization and Section 4.3 the normalization. Section 4.4 documents the order of the steps, the software and two defects found and corrected during this work, and Section 4.5 summarizes what the preprocessing achieved.

4.1 Data exploration

4.1.1 Structure and completeness

The exploration concentrated on the MEDA 5-minute table, which feeds the single-site experiments, and compared the three stations on the hourly tables (Section 4.2.4). The MEDA table has 255,132 rows and 19 columns, all of them numerical, with one row per 5-minute interval of the spacecraft clock and no duplicate interval. It covers 1,473 of the 1,619 sols. Between its first and its last interval, the clock spans 478,890 intervals, of which 53.3 % hold at least one value. The other intervals form 21,361 gaps with a median length of 45 minutes. Only 364 gaps last 15 minutes or less, and 442 last an hour or more, so most gaps separate the measurement sessions of a sol rather than interrupt them. Because the sessions alternate between hours (Chapter 3), no hour of the sol is left out: between 50 and 64 % of the intervals of every local mean solar time (LMST) hour hold data in the sols with measurements.

Completeness differs strongly between the variables (Table 4.1). Air temperature is present in

[1] Cf. J. W. Tukey. *Exploratory Data Analysis*. Reading, MA: Addison-Wesley, 1977.

Table 4.1: Completeness and descriptive statistics of the MEDA 5-minute table

Variable	Unit	Present (%)	Mean	Median	SD	Skewness	Excess kurtosis	Beyond fences (%)	Maximum
`T_air`	K	99.8	222.6	218.1	20.0	0.34	−1.18	0.00	271.6
`pressure`	Pa	94.9	704.5	711.5	53.3	−0.29	−0.94	0.00	820.4
`wind_speed`	$\mathrm{m\,s^{-1}}$	11.6	3.40	3.09	1.98	0.72	0.04	0.83	13.1
`vmr_h2o`	ppm	31.4	41.3	35.0	72.0	58.6	4,545	5.15	7,508
`T_humidity_local`	K	89.2	226.3	222.5	21.2	0.20	−1.38	0.00	277.9
`LW_down`	$\mathrm{W\,m^{-2}}$	94.2	32.3	31.0	8.3	1.28	2.93	2.85	98.8
`LW_up`	$\mathrm{W\,m^{-2}}$	94.2	168.7	122.1	204.9	28.2	1,169	0.81	11,332
`RDS_TOP_avg`	$\mathrm{W\,m^{-2}}$	53.0	18.1	17.5	13.2	0.29	−1.02	0.00	55.9
`RH` (raw)	%	89.1	13.1	0.45	983	291	103,540	11.09	380,304

SD: standard deviation. Fences: first quartile minus and third quartile plus 1.5 interquartile ranges.
Source: own elaboration based on the MEDA 5-minute table (Chapter 3).

99.8 % of the rows, pressure in 94.9 % and the thermal infrared fluxes in 94.2 %. Solar radiation is present in 53.0 % of the rows, the water-vapor mixing ratio in 31.4 % and wind in 11.6 %, only in sols 36–312. The mixing ratio is a night-time variable: it is present in 85–91 % of the intervals between 0 and 6 LMST but in at most 3 % of those between 9 and 21 LMST. All variables of the table are continuous. The only categorical variable of the project, the name of the station, appears in the hourly tables, and the quality flags of the archives are text (Chapter 3).

4.1.2 Distributions

Table 4.1 summarizes the variables, and Figure 4.1 shows the distributions of the three forecast targets. Air temperature is bimodal, with a night-time mode near 202 K and a day-time mode near 248 K, so most of its standard deviation of 20 K is the diurnal cycle. Pressure has three modes between 600 and 800 Pa. They are not separate populations: the seasonal cycle of pressure dwells longer near some levels than near others (Section 4.3.3). Wind speed is skewed to the right, with a median of 3.1 $\mathrm{m\,s^{-1}}$ and a skewness of 0.72. None of the three targets is strongly skewed, so no transformation such as a logarithm or a Box–Cox transformation[2] was applied, and all targets are kept in their physical units.

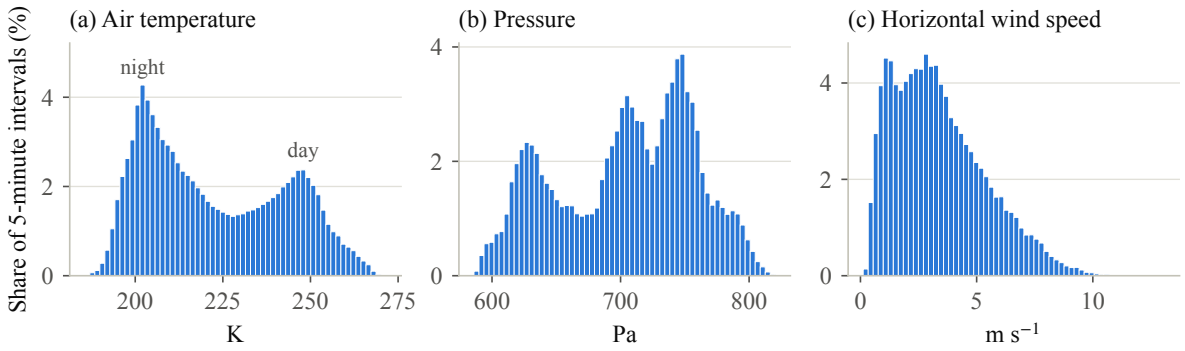
4.1.3 Correlations

Relations between the variables were measured with Spearman's rank correlation coefficient[3], the Pearson correlation of the ranks of the values. It captures any monotone relation and is insensitive to extreme values. Between air temperature and the upward longwave flux, for example, the Pearson coefficient is 0.41 and the Spearman coefficient 0.93; the difference is due to the extreme

[2]Cf. R. J. Hyndman and G. Athanasopoulos. *Forecasting: Principles and Practice*. 3rd ed. Melbourne: OTexts, 2021. URL: `https://otexts.com/fpp3/` (accessed 2026-10-07).

[3]C. Spearman. "The Proof and Measurement of Association between Two Things". In: *The American Journal of Psychology* 15.1 (1904), pp. 72–101.

Figure 4.1: Distributions of the forecast targets at Jezero Crater



Source: own elaboration based on the MEDA 5-minute table (Chapter 3).

tail of the flux (Section 4.1.4). Each coefficient uses the rows in which both variables are present, so the coefficients of wind refer to sols 36–312 and those of the mixing ratio mainly to the night.

Figure 4.2 shows three groups. First, air temperature, the temperature of the humidity sensor, both longwave fluxes and solar radiation all follow the diurnal heating and are positively related: air temperature, the humidity-sensor temperature and the upward flux closely ($\rho = 0.93$–0.98), the downward flux and solar radiation more loosely ($\rho = 0.21$–0.82). Second, pressure is almost unrelated to this group ($|\rho| \leq 0.24$) but anticorrelated with the mixing ratio ($\rho = -0.62$). The relation holds between daily means ($\rho = -0.62$) but nearly vanishes within sols ($\rho = -0.10$), so it reflects a shared seasonal cycle rather than a coupling at short time scales. Third, the raw relative humidity is anticorrelated with air temperature ($\rho = -0.91$) because it refers to the temperature of the sensor head, not to that of the air (Chapter 3). Because several secondary variables are closely related to each other, they are added to the models as one block rather than one at a time (Chapter 6).

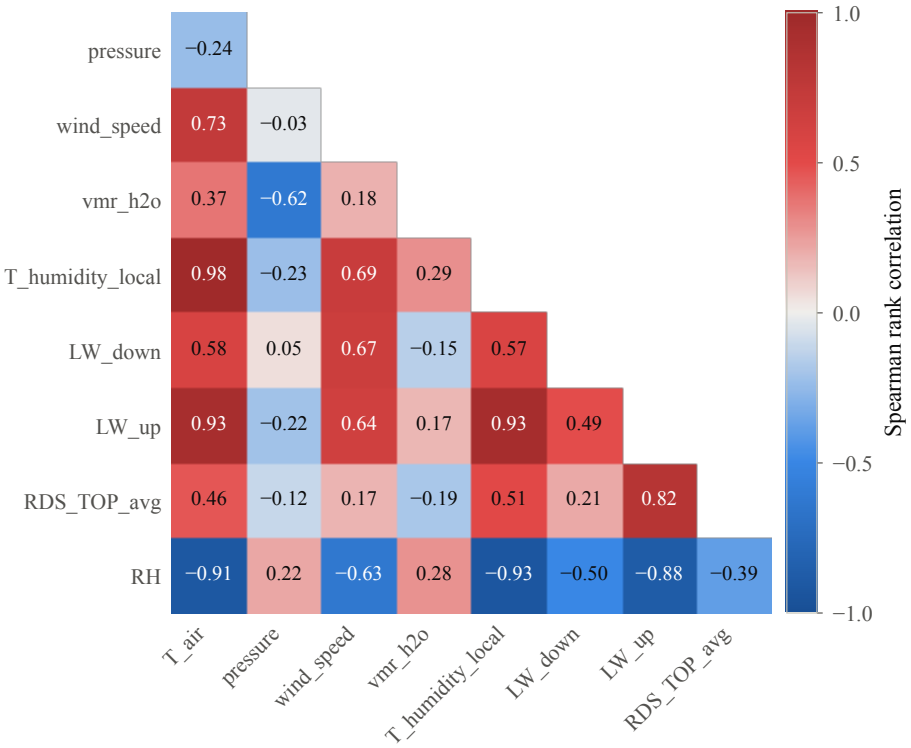
4.1.4 Outliers and noise

Outliers were screened with Tukey's fences, which mark values lying more than 1.5 interquartile ranges below the first or above the third quartile[4], and with the largest absolute z-score. The decision rule was physical. Extreme values that the atmosphere can produce were kept, because rare events such as gusts and dust storms are part of what a forecaster must handle. Values that a sensor cannot have measured correctly were removed by the quality control (Chapter 3) or, in model inputs, limited (Section 4.3.2).

Air temperature and pressure have no values beyond the fences, and their largest absolute z-scores are 2.4 and 2.2. Wind speed has 0.8 % of its values beyond the fences, up to 13.1 $\mathrm{m\,s^{-1}}$; these are gusts, and they were kept. Three secondary variables have heavy tails (Table 4.1). The raw relative humidity reaches 3.8×10^5 % and serves only as a diagnostic (Chapter 3). The mixing ratio exceeds 200 ppm in 83 intervals (0.1 %), mostly in the afternoon and at times when the raw relative humidity is also extreme (median 4,700 %); only the tree model uses this

[4]Cf. Tukey, *op. cit.*

Figure 4.2: Spearman rank correlations between the MEDA variables



Source: own elaboration based on the MEDA 5-minute table (Chapter 3).

variable, and it was left as it is (Section 4.3.1). The upward longwave flux is the clearest case. Its largest value in the training period is 423 W m^{-2}, but between sols 1551 and 1601, 1,596 values exceed 500 W m^{-2}, which a black body would emit only above 306 K, and the maximum is 11,332 W m^{-2}. These values make up 4.0 % of the flux in the test period. They are treated as instrument artifacts.

4.2 Data standardization

4.2.1 Variables, units and missing values

The archives already report every physical quantity in SI units or, for the mixing ratio, in parts per million, so no unit had to be converted; angles are converted from degrees to radians only inside trigonometric functions. Standardization therefore concerned names, missing values, time and sampling. Table 4.2 maps the source columns of each station to the common variable names. The three missing-value conventions, empty cells and 999999999 in MEDA, −999 in REMS and empty cells in APSS, were converted to the single not-a-number value of NumPy and pandas[5], which Parquet stores as null. Time fields were parsed from text into numbers, and the flag strings were evaluated by the quality-control rules and not kept in the derived tables.

[5]C. R. Harris et al. “Array programming with NumPy”. In: *Nature* 585.7825 (2020), pp. 357–362; W. McKinney. “Data Structures for Statistical Computing in Python”. In: *Proceedings of the 9th Python in Science Conference*. 2010, pp. 56–61.

Table 4.2: Harmonization of the three station records

Common variable	MEDA	REMS	APSS
`T_air` (K)	mean of `ATS_LOCAL_TEMP1–5`	`AMBIENT_TEMP`	mean of `BMY_AIR_TEMP` and `BPY_AIR_TEMP`
`pressure` (Pa)	`PRESSURE`	`PRESSURE`	`PRESSURE`
`wind_speed`, `wind_u`, `wind_v` ($\mathrm{m\,s^{-1}}$)	`HORIZONTAL_WIND_SPEED`, `WIND_DIRECTION`	same columns; no value passes quality control	`HORIZONTAL_WIND_SPEED`, `WIND_DIRECTION`
Missing value	empty cell, 999999999	−999	empty cell
Time keys	`SCLK`, `LMST`	`TIMESTAMP` (spacecraft clock), `LMST`	`SCLK`, `LMST`
Sampling	1–2 Hz in sessions of about 5 minutes to one hour	1 Hz, 5 minutes per hour and longer blocks	0.1–1 Hz (air temperature, wind), 2–20 Hz (pressure)
5-minute grid	mean of the samples per interval	—	—
Hourly grid	mean of the 5-minute values, at least one interval	mean of the samples, at least 60	mean of the samples, at least 60

Source: own elaboration based on the labels and format files of the archives[6].

4.2.2 Time base

The archives record time in several forms (Chapter 3), of which the pipeline uses three. It parses LMST strings such as `00300M02:00:30.032` into a sol number and a fractional hour, here 300 and 2.008 h, and uses them to place every sample in its sol and hour; for MEDA, the sol is taken from the daily file of the pressure product instead (Section 4.2.3). Coordinated Universal Time (UTC) is estimated as the spacecraft clock plus an epoch fixed on one product of each mission, and the solar longitude L_s of the hourly tables is computed from UTC with the Mars24 algorithm[7]. The MEDA feature table uses a simpler seasonal coordinate, computed from the sol

[6] J. A. Rodriguez-Manfredi et al. *The Mars Environmental Dynamics Analyzer, MEDA*. Data bundles of the Mars 2020 mission. NASA Planetary Data System, Atmospheres Node. 2021. URL: `https://pds-atmospheres.nmsu.edu/data_and_services/atmospheres_data/PERSEVERANCE/meda.html` (accessed 2026-10-07). DOI: 10.17189/1522849; J. Gomez-Elvira. *Mars Science Laboratory Rover Environmental Monitoring Station RDR Data V1.0*. MSL-M-REMS-5-MODRDR-V1.0. NASA Planetary Data System, 2013. URL: `https://pds.nasa.gov/ds-view/pds/viewDataset.jsp?dsid=MSL-M-REMS-5-MODRDR-V1.0` (accessed 2026-10-18); J. A. Rodriguez-Manfredi. *APSS TWINS Data*. NASA Planetary Data System, 2019. URL: `https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:insight_twins` (accessed 2026-10-18); D. Banfield. *APSS PS Data*. NASA Planetary Data System, 2019. URL: `https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn:nasa:pds:insight_ps` (accessed 2026-10-18).

[7] M. Allison and M. McEwen. "A Post-Pathfinder Evaluation of Areocentric Solar Coordinates with Improved Timing Recipes for Mars Seasonal/Diurnal Climate Studies". In: *Planetary and Space Science* 48.2–3 (Feb. 2000), pp. 215–235.

number alone:

$$\tilde{L}_s = \left(5.5^\circ + 360^\circ \cdot \frac{\text{sol}}{668.6}\right) \bmod 360^\circ, \tag{4.1}$$

where 5.5° approximates the solar longitude at landing and 668.6 is the length of a Mars year in sols. Because Mars moves along its eccentric orbit at a varying speed, $\tilde{L}_s$ differs from L_s by up to 21°, and by about 10° on average over sols 1–1619. It repeats with the Mars year, so it is a consistent seasonal coordinate for the calendar features and the seasonal baseline of pressure (Section 4.3.3), but it is not the solar longitude. Forecast horizons are counted in Earth hours on the 5-minute grid, which follows the spacecraft clock, and in Mars hours of 3,699 s on the hourly grid.

4.2.3 Temporal aggregation and integration

Aligning the samples in time was the main integration step, both between the seven MEDA products and between the three stations. For MEDA, the samples of each product, recorded at 1 Hz (2 Hz for some products in the first 150 sols), were averaged over 5-minute intervals of the spacecraft clock, with the interval index $\lfloor \text{SCLK}/300 \rfloor$. Intervals split between two daily files were averaged, the local time on the 24-hour circle so that such an interval stays at midnight, and the seven products were joined on the interval index. Air temperature is the mean of the five ATS units and solar radiation the mean of the eight top photodiodes of the radiation sensor. The sol of each interval is taken from the pressure product and its local time from the ancillary product.

For all three stations, the hourly tables share one grid of LMST hours. The valid samples of each sol and LMST hour were averaged, and a mean was kept only when enough samples supported it (Table 4.2); the number of samples is stored with it. For REMS and APSS, means of the same hour from different files were combined with weights equal to their sample counts. Wind direction was never averaged directly: the speed and direction of every sample were first converted to the components u and v, which were then averaged (Section 4.3.4). Finally, both grids were completed. The MEDA table was reindexed on all 478,890 intervals and each hourly table on every Mars hour between its first and its last row. Missing intervals and hours became empty rows, so that a lag of one hour or a 24-hour window refers to time rather than to a number of rows.

4.2.4 Harmonized records of the three stations

Figure 4.4 compares the stations on the hourly grid over the training period of each site, as defined for the cross-site experiments (Chapter 6). Their levels differ. The median air temperature is 217 K at Jezero, 219 K at Gale and 200 K at Elysium Planitia, and the median pressure 731, 838 and 721 Pa. The mean diurnal cycle, computed as the deviation of each hourly value from the mean of its sol, has nearly the same timing at all sites: the mean cycles of any two sites correlate with $r = 0.93$–0.99 for air temperature and $r = 0.86$–0.96 for pressure. Its amplitude, however, is local. The diurnal range is 55.9, 51.7 and 69.5 K for air temperature and 24.4, 75.5 and 39.6 Pa

for pressure. The large pressure range at Gale agrees with the hydrostatic adjustment of the air over the topography of the crater region that has been simulated there[8]. These differences shaped the normalization of the cross-site experiments (Section 4.3.5).

4.3 Data normalization

4.3.1 Choice of technique for each model

Two common techniques rescale a variable x to a common scale. Min–max normalization maps the training minimum and maximum to 0 and 1,

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \tag{4.2}$$

and z-score normalization subtracts the mean μ and divides by the standard deviation σ,

$$z = \frac{x - \mu}{\sigma}. \tag{4.3}$$

A third technique, L2 normalization, scales vectors to unit length and suits feature vectors such as text embeddings, which do not occur here. Whether to rescale at all depends on the model. Decision trees, and hence gradient-boosted decision trees (GBDT), split on the order of the values of each input, so they are invariant under strictly monotone transformations of the inputs[9]. The GBDT therefore received its inputs in physical units, which also keeps its split thresholds interpretable. A neural network trained by gradient descent, by contrast, converges faster when its inputs have means close to zero and similar variances[10], so the inputs of the temporal convolutional network (TCN) were z-scored. Min–max normalization was rejected because its scale is set by the two most extreme values, which the exploration showed to be unreliable (Section 4.1.4), and because test values can fall outside the training range: 4.1 % of the pressures of the test period lie below the training minimum.

For each input channel of the TCN, μ and σ were computed from the observed values at the training issue times, the times at which the training forecasts are issued, and applied unchanged to every period. Missing inputs were set to zero after scaling, that is, to the training mean, and five binary flags, which are not scaled, mark where the pressure, the wind speed, the mixing ratio, the downward longwave flux and the solar radiation are missing. The forecast target was z-scored in the same way and the predictions were transformed back, so that all errors are reported in physical units.

[8] Cf. M. I. Richardson and C. E. Newman. “On the relationship between surface pressure, terrain elevation, and air temperature. Part I: The large diurnal surface pressure range at Gale Crater, Mars and its origin due to lateral hydrostatic adjustment”. In: *Planetary and Space Science* 164 (2018), pp. 132–157.

[9] Cf. T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd ed. Springer Series in Statistics. New York: Springer, 2009.

[10] Cf. Y. A. LeCun et al. “Efficient BackProp”. In: *Neural Networks: Tricks of the Trade*. Ed. by G. Montavon, G. B. Orr, and K.-R. Müller. 2nd ed. Lecture Notes in Computer Science 7700. Berlin, Heidelberg: Springer, 2012, pp. 9–48.

4.3.2 Limiting extreme values

Apart from the diagnostic relative humidity and a few values of the mixing ratio, which only the tree model with secondary signals uses, the only variable with values that no sensor reading can justify is the upward longwave flux (Section 4.1.4). Its values were capped at the 99.5th percentile of the training period, $q_{0.995} = 372.4\ \mathrm{W\,m^{-2}}$:

$$x' = \min\left(x,\ q_{0.995}\right). \tag{4.4}$$

The cap is a percentile rather than a fixed physical limit, so it is estimated from the training data alone. It reduces the excess kurtosis of the flux from 1,169 to −0.55 and its skewness from 28.2 to 0.84. It changes 0.5 % of the training values but 6.5 % of the test values; besides the artifacts, these include about 2.4 % of the test values that are plausible but lie above the cap. The flux is an input only of the models with secondary signals, and no forecast target is capped.

4.3.3 Seasonal baseline of pressure

Absolute pressure is dominated by the seasonal cycle. A seasonal baseline b was therefore estimated as the median pressure of the training period in each of 36 bins of $\tilde{L}_s$, 10° or 18.6 sols wide, and the deviation

$$p' = p - b(\tilde{L}_s) \tag{4.5}$$

was formed. The baseline ranges from 620 to 790 Pa, and subtracting it reduces the standard deviation of pressure in the training period from 55.3 to 11.5 Pa (Figure 4.3). The deviation, its lags and its windowed statistics are inputs of the GBDT, the baseline itself is an input of both the GBDT and the TCN, and the deviation is the target of the seasonal-naive reference forecast (Chapter 6). Removing a seasonal component in this way is common practice in forecasting[11].

The baseline does not remove all slow variation. The median deviation is 0.0 Pa in the training period but −28.9 Pa in the validation period and −36.1 Pa in the test period (Figure 4.3b). The season cannot explain this shift, but the rover's position can. The ancillary MEDA product reports the position of the rover[12], and its vertical coordinate, read here as the height above the landing point, rose from a median of 46 m in the training period to 631 m in the test period. During the validation period, Perseverance climbed about 500 m to the rim of Jezero Crater[13], from 197 m at its start to 709 m at its end. In an atmosphere in hydrostatic balance, pressure decreases with height z as

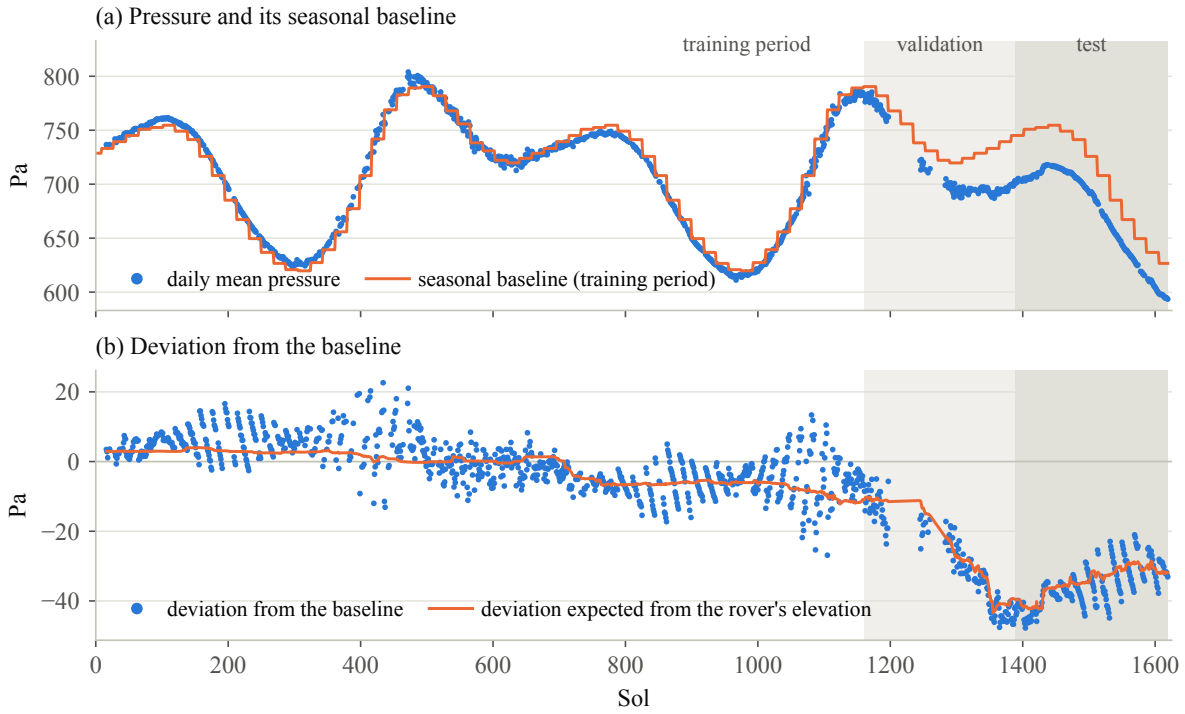
$$p(z) = p(z_0)\ \exp\!\left(-\frac{z - z_0}{H_p}\right), \qquad H_p = \frac{R\,T}{g}, \tag{4.6}$$

[11] Cf. Hyndman and Athanasopoulos, *op. cit.*

[12] Rodriguez-Manfredi et al., *op. cit.*, labels of the ANCILLARY products.

[13] NASA Jet Propulsion Laboratory. *NASA's Perseverance Rover Reaches Top of Jezero Crater Rim.* News release. Dec. 12, 2024. URL: `https://www.jpl.nasa.gov/news/nasas-perseverance-rover-reaches-top-of-jezero-crater-rim/` (accessed 2026-10-26).

Figure 4.3: Daily mean pressure at Jezero Crater before and after removal of the seasonal baseline



Source: own elaboration based on the MEDA 5-minute table and the ancillary products (Chapter 3).

where the pressure scale height H_p depends on the specific gas constant R of the air, its temperature T and the gravity g. With the mean molecular weight of Martian air and the surface gravity of Mars[14] and the daily mean air temperature, $H_p \approx 11.4$ km, and the expected gradient near the surface is $-0.0624\ \mathrm{Pa\,m^{-1}}$. The daily mean deviations follow the elevation of the rover with a slope of $-0.0627\ \mathrm{Pa\,m^{-1}}$ ($r = -0.95$), and once the hydrostatic offset expected from the elevation is removed, the median deviation of every period lies within 2 Pa of zero. The shift of the later periods is therefore an effect of the rover's elevation, not of the weather. The models use the pressure as the instrument measured it at the rover, so they face this shift in the test period; its consequences are discussed in Chapter 7.

4.3.4 Encoding of periodic variables

Periodic variables were encoded by the sine and cosine of their phase, so that the end and the start of a cycle lie next to each other: the LMST hour h as $(\sin 2\pi h/24,\ \cos 2\pi h/24)$ and the season as $(\sin \tilde{L}_s,\ \cos \tilde{L}_s)$ on the 5-minute grid or $(\sin L_s,\ \cos L_s)$ on the hourly grid. Wind, given as a speed s and a direction θ, was converted to the components

$$u = s\cos\theta, \qquad v = s\sin\theta, \tag{4.7}$$

[14] D. R. Williams. *Mars Fact Sheet*. NASA Goddard Space Flight Center. May 19, 2025. URL: `https://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html` (accessed 2026-10-26).

which removes the jump between 359° and 0° and allows directions to be averaged as vectors. These encodings need no fitted parameters.

4.3.5 Relative inputs for the cross-site experiments

A model transferred to another site meets a different level of every variable (Section 4.2.4). In the cross-site experiments, every input and the target were therefore expressed relative to the current state of the series v:

$$\Delta_k v(t) = v(t) - v(t-k), \qquad d_w v(t) = v(t) - \frac{1}{w}\sum_{j=0}^{w-1} v(t-j), \tag{4.8}$$

with lags $k \in \{1, 2, 3, 6, 12, 24, 48\}$ h and windows $w \in \{6, 24\}$ h, together with the standard deviation over each window and the encodings of the hour and the season. The target is the change $y = v(t+H) - v(t)$ over the forecast horizon H. Differencing of this kind removes the level of a series[15]. Figure 4.4 shows its effect. The distributions of the absolute values of the three sites are displaced by up to 117 Pa and 20 K, whereas the one-hour changes of all sites are centered at zero, with means within 0.07 Pa and 0.02 K. Their spread still differs: the standard deviation of the one-hour pressure change is 5.1 Pa at Jezero, 9.1 Pa at Gale and 5.7 Pa at Elysium Planitia. This difference in amplitude was deliberately not normalized away. Scaling each site by its own statistics would require data from the target site, which a transferred model is assumed not to have, and the amplitude difference is part of what the cross-site experiments measure.

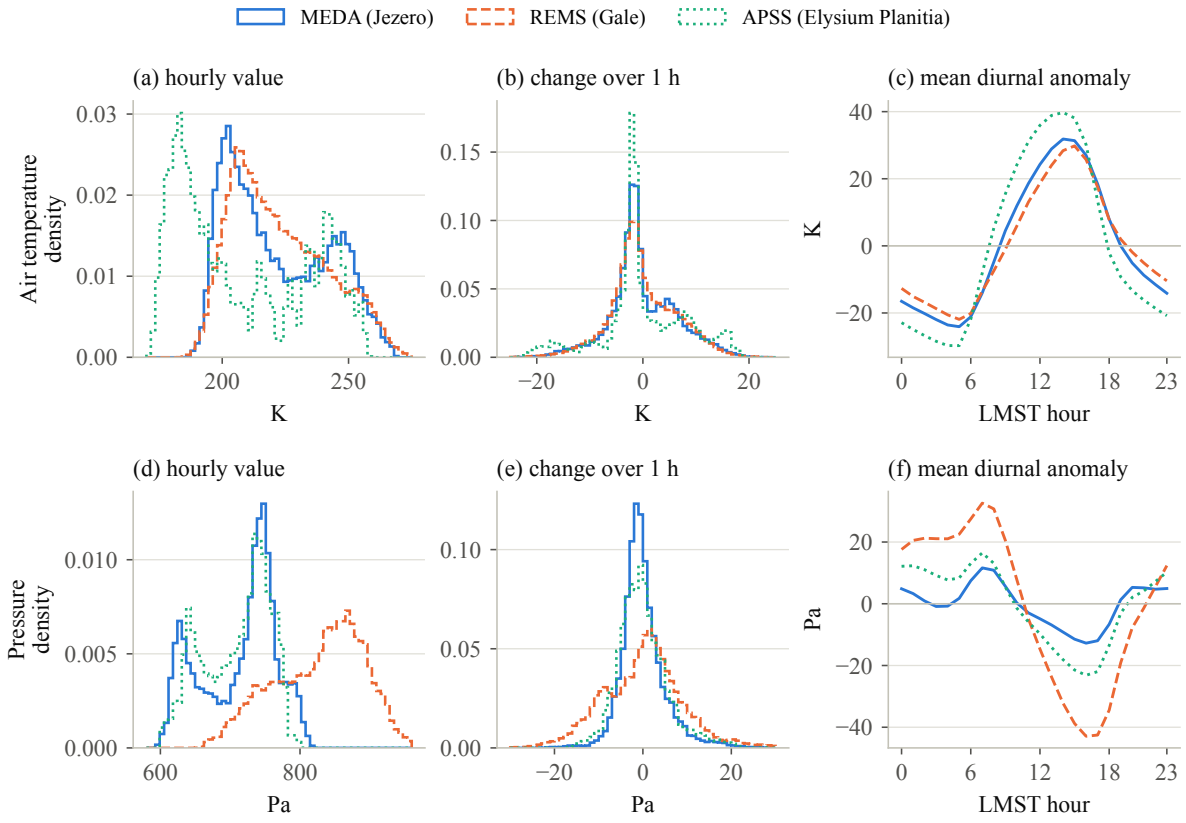
The rows of the cross-site experiments are also complete: a forecast issue time is used only when all its inputs and its target are present. The duty cycles of the instruments differ between sites and over time, and a pattern of missing inputs that occurs at only one site would confound the comparison. In a preliminary version, which gave each model the history of both variables, allowed missing inputs and divided the periods by the number of sols alone, the model trained and tested at Elysium Planitia forecast air temperature six hours ahead with an error of 13.6 K, worse than the 6.6 K of a time-of-sol climatology. With univariate inputs, complete rows and period boundaries placed over well-covered sols (Chapter 6), the same error fell to 2.9 K.

4.3.6 Missing values and leakage control

Missing values were treated differently in inputs and in targets (Table 4.3). In the MEDA tables, gaps of at most three intervals (15 minutes) in an input were bridged with the last observed value, and longer gaps stay missing as a whole. This bridges 610 values of air temperature, 0.24 % of those measured; short gaps in pressure remain, because the seasonal baseline to which pressure is referred (Section 4.3.3) needs the sol number, which comes from the pressure product itself. Whether a gap is short is known only once it has ended, so the rule uses information that a forecaster working in real time would not have; it affects none of the air-temperature test forecasts

[15] Cf. Hyndman and Athanasopoulos, *op. cit.*

Figure 4.4: Hourly air temperature and pressure at the three stations in absolute and relative form



Source: own elaboration based on the hourly tables (Chapter 3); training periods of the cross-site experiments.

and 1–17 of the 1,619–2,796 wind test forecasts per horizon. The missingness flags are computed before any filling. The GBDT receives missing inputs as they are, because LightGBM handles missing values natively[16], and the TCN receives them as the training mean together with the flags (Section 4.3.1). A target, in contrast, is never filled: a forecast is scored only where the target was measured, and rows without a measured target are removed.

Every statistic estimated from the data, namely the cap of the longwave flux, the seasonal baseline and the means and standard deviations of the TCN, was estimated on the training period only and then applied unchanged to the validation and test periods, and all lags and windows use present and past values only. This prevents leakage, the use of information in training that would not be available when a forecast is made[17]. The training, validation and test periods themselves are defined in Chapter 6.

[16]Microsoft Corporation. *Advanced Topics*. LightGBM 4.3.0 documentation. 2024. URL: `https://lightgbm.readthedocs.io/en/v4.3.0/Advanced-Topics.html` (accessed 2026-10-26).

[17]Cf. S. Kaufman et al. “Leakage in Data Mining: Formulation, Detection, and Avoidance”. In: *ACM Transactions on Knowledge Discovery from Data* 6.4 (2012), pp. 1–21.

Table 4.3: Normalization, encoding and missing-value treatment by model

Technique	Applied to	Estimated on	Used by
none (physical units)	all inputs	—	GBDT
z-score, Eq. (4.3)	inputs and target	training issue times	TCN
cap at $q_{0.995}$, Eq. (4.4)	upward longwave flux	training period	GBDT with secondary signals
seasonal baseline, Eq. (4.5)	pressure	training period	GBDT and TCN (inputs), seasonal-naive forecast (target)
sine and cosine of the phase	LMST hour, season	—	GBDT, TCN, cross-site GBDT
wind components, Eq. (4.7)	wind	—	GBDT, TCN
changes and deviations, Eq. (4.8)	hourly air temperature and pressure	—	cross-site GBDT
last value carried forward	input gaps of $\leq$ 15 minutes	—	GBDT, TCN
missingness flags	five MEDA sensors	—	GBDT, TCN
measured targets only	all targets	—	all models
complete rows	inputs and target	—	cross-site GBDT

Source: own elaboration.

4.4 Reproducibility

Figure 4.5 gives, in pseudocode, the order of the steps that turn the MEDA 5-minute table into the training data of a model. The order follows one rule: no statistic is estimated from the data before the training period is known. Each step is implemented in the project code (Chapter 5); the inputs and the main settings come from configuration files, while a few constants, such as the 15-minute limit of the gap filling and the 36 bins of the seasonal baseline, are fixed in the code. Every run writes its resolved configuration, a log and a summary to its own report folder. The figures of this chapter are drawn by a script stored next to them, which reads only the tables named in their source lines and results derived from them.

Table 4.4 lists the software and its versions. The exploratory statistics and the figures were produced with pandas, NumPy and Matplotlib[18], the GBDT with LightGBM[19] and the TCN with PyTorch[20]. The MEDA 5-minute table was built on a Windows laptop with Python 3.11.5, and the other tables and the final experiments on a MacBook with an Apple M4 processor and Python 3.12.14, with the same versions of all other packages.

Four defects of the preprocessing code were found during this work and corrected before the

[18]McKinney, *op. cit.*; Harris et al., *op. cit.*; J. D. Hunter. "Matplotlib: A 2D Graphics Environment". In: *Computing in Science & Engineering* 9.3 (2007), pp. 90–95.

[19]G. Ke et al. "LightGBM: A Highly Efficient Gradient Boosting Decision Tree". In: *Advances in Neural Information Processing Systems*. Vol. 30. Curran Associates, Inc., 2017, pp. 3146–3154. URL: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html (accessed 2026-10-07).

[20]A. Paszke et al. "PyTorch: An Imperative Style, High-Performance Deep Learning Library". In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc., 2019. URL: https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html (accessed 2026-10-07).

Figure 4.5: Order of the preprocessing steps for the MEDA single-site experiments

```
input : MEDA 5-minute table T, configuration of the feature set
 1  F <- T reindexed on every 5-minute interval of the clock
 2  F <- flags marking where each of five sensor variables is missing
 3  F <- wind components u, v; sine/cosine of LMST hour and L~s
 4  F <- marks of the measured values of each target
 5  F <- inputs with gaps of <= 3 intervals filled with the last value
 6  F <- training/validation/test labels by sol (Chapter 6)
 7  cap <- 99.5th percentile of LW_up in the training rows
    F.LW_up <- min(F.LW_up, cap)
 8  b <- median pressure per bin of L~s in the training rows
    F.pressure_dev <- F.pressure - b(L~s)
 9  F <- lags (5, 60, 360, 1440 min) and window means and standard
          deviations (60, 360, 1440 min) of the inputs
10  for each target y and horizon H:
        Y(t) <- y(t + H), kept only where y(t + H) was measured
        if model = TCN: mu, sigma <- statistics of the training rows;
                        z-score inputs and Y; missing inputs <- 0
        fit on the training rows, stop early on the validation rows
```

Source: own elaboration based on the project code.

final experiments. First, the step meant to bridge only gaps of up to 15 minutes also filled the first 15 minutes of every longer gap, because the fill was limited in length but not restricted to short gaps. It had added 63,607 copied values of air temperature to the feature table, a quarter of the measured ones, and 4–23 % of the air-temperature targets and 11–30 % of the wind targets of the test period, depending on the horizon, had been such copies. Second, the TCN had received missing inputs as zeros in physical units, which after scaling lay 11–13 standard deviations below the mean in the models for air temperature and pressure and up to 29 in those for wind. Third, the MEDA wind direction had been averaged over each 5-minute interval as a plain number, which shifted the components u and v by more than 1 $\mathrm{m\,s^{-1}}$ in 13.8 % of the intervals with wind. Fourth, the local time of the 610 intervals at local midnight that were split between two daily files had been averaged to noon. In addition, the duplicate intervals of the 5-minute table had been merged by a step outside the project code; this step is now part of it. The corrected code fills only short gaps, scores forecasts only against measured targets, scales the TCN inputs before filling them (Section 4.3.1) and averages wind as a vector and local time on the circle (Section 4.2.3). Automated tests check each correction, the changes are recorded with the project, and every experiment reported in Chapter 7, including the search for the TCN configuration, is run on the corrected data.

Table 4.4: Software used for the preprocessing

Package	Version	Use
Python	3.11.5, 3.12.14	all code (Windows, macOS)
NumPy	1.26.4	arrays and numerical operations
pandas	2.2.2	tables, aggregation, statistics
PyArrow	15.0.2	reading and writing Parquet files
LightGBM	4.3.0	GBDT
PyTorch	2.2.2	TCN
Matplotlib	3.8.4	figures
PyYAML	6.0.1	configuration files
pytest	8.2.0	automated tests of the preprocessing

Source: own elaboration.

4.5 Summary

Preprocessing turned three heterogeneous archives into tables in which every row is a point in time on a common grid, every variable has one name, one unit and one representation of missing values, and every gap is visible. The exploration showed what the models face: a diurnal cycle that dominates air temperature, a seasonal cycle and a drift with the rover's elevation that dominate pressure, closely related secondary variables, sparse wind and humidity data, instrument artifacts in the upward longwave flux, and site differences in level and amplitude. Normalization answered these findings model by model. The trees received physical units and the TCN z-scored inputs; pressure was referred to a seasonal baseline; periodic variables received sine and cosine encodings; and the cross-site experiments received relative inputs and targets that remove the level of each site.

The preprocessing measurably improved the data and the models. The corrected gap filling removed about 63,000 copied values of air temperature from the feature table, so every forecast is now scored against a measurement, and averaging the wind as a vector corrected the wind components of 13.8 % of the intervals with wind. In the cross-site experiments, the redesigned inputs and periods reduced the six-hour error of the model for air temperature at Elysium Planitia from 13.6 to 2.9 K. Two limitations remain and are carried into the following chapters. The pressure measured by the climbing rover lies about 36 Pa below the seasonal baseline in the test period, a shift that the baseline cannot capture, and the cap on the upward longwave flux removes its artifacts only together with some plausible values.

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